

ABARES undertakes a range of multi-disciplinary research projects that assist the department and external clients to achieve strategic objectives as they relate to biosecurity issues in invasive species, social sciences and quantitative sciences.

This includes

- improving techniques for mapping and modelling pest and disease spread
- developing methods for designing more efficient surveillance systems
- developing processes to assist with the prioritisation of exotic invasive species
- providing advice on biosecurity policy and proposed reforms
- assessing tools and mechanisms to achieve effective community engagement in biosecurity.

Research reports

Published: 25 February 2026

This report provides new mathematical modelling techniques to support the design and monitoring of compliance-based biosecurity intervention systems at the Australian border, such as the Compliance-Based Incentive Scheme (CBIS). These systems use a risk-based intervention approach to manage imports while reducing regulatory costs.

Published: 19 August 2025

Asian green mussel has been detected on vessels arriving in Australia but has not become established. Results from this analysis show that it could potentially become established in the northern half of Australia if introduced.

ABARES used the Species Range Mapping modelling method to generate a map showing its maximum potential range based on its known temperature tolerances.

Published: 25 July 2025

Climatch is a software tool widely used for assessing the establishment risk of invasive species. This report complements the existing Climatch user manual by offering an in-depth explanation of the two algorithms Climatch employs to assess climate suitability.

Published: 7 March 2025

Amur River clam is a species of bivalve mollusc that is native to the north-western Pacific Ocean, including eastern China, Korea, Japan and far eastern Russia. It came to prominence when an incursion was discovered in San Francisco Bay on the west coast of the USA in the late 1980s where it achieved densities of more than 10,000 individuals/m². An incursion has also occurred in the estuary of the River Scheldt in Belgium where it was first detected in 2018.

Amur River clam has not been detected in Australia but the results from this analysis show that it could potentially become established in the southern half of Australia if introduced.

Published: 1 August 2024

Suminoe oyster is a large true oyster that is native to China but has been spread to many countries in Asia, including, Japan, South Korea and India. Its likely impacts include outcompeting native species and it may harbour parasites and diseases. It is a known invasive species that has invaded localities across Asia.

It was found in Moreton Bay in Queensland for the first time in Australia in June 2023. ABARES used the Species Range Mapping modelling method to generate a map showing its maximum potential range based on its known temperature tolerances.

Published: 13 June 2024

Carpet sea squirt is a colonial ascidian that is thought to be native to Japan. It is a filter feeder that can overgrow everything and spread over large areas. It is a known invasive species that has invaded localities in many places around the world.

It was found in Western Australia for the first time in Australia in April 2020 and has since spread to Sydney. ABARES used the Species Range Mapping modelling method to generate a map showing its maximum potential range based on its known temperature tolerances.

Published: 30 April 2021

National coordination of weeds data is a priority for all levels of government in Australia and is established in national policy documents such as the Australian Weeds Strategy (AWS) 2017–2027, the Intergovernmental Agreement on Biosecurity (IGAB), and the national strategies for Weeds of National Significance (WoNS). ABARES conducted a national survey of state, territory and Natural Resource Management (NRM) institutional stakeholders to understand the status of weeds data collection and sharing, and identify opportunities for national data coordination. This report presents results of responses from over two thirds of NRM regions and from state and territory agencies representing all jurisdictions.

Key issues

The national weeds data collection survey provided an understanding of the state of weeds data in Australia, including for nationally listed Weeds of National Significance, National Environmental Alert weeds and Agricultural Sleeper Weeds. Responses to the survey indicate that a range of data on established and emerging weeds are being collected, maintained and used extensively by different agencies for their region. Both state and territory and NRM respondents reported availability of funding and human resources as key barriers to collecting weeds data.

Published: 25 June 2020

General surveillance is increasingly seen as a cost-effective way to obtain monitoring data about pest and disease status. Different forms of general surveillance are already making a considerable contribution to Australia's biosecurity system. However general surveillance can be challenging to instigate and maintain due to interrelated social, institutional, organisational, ecological and

infrastructure factors, and the fact that general surveillance is plagued by fragmentation. This report explores general surveillance from a broad perspective, compiling lessons learned from recent literature to capture key considerations for the different components of general surveillance and the dynamics between them.

Key Issues

This report offers insights about the different components of general surveillance initiatives comprising (i) actors and their relationships; (ii) infrastructure (physical, knowledge and financial systems); (iii) institutions (formal and informal rules); and (iv) the biophysical components. It illustrates how the interactions within components, among them and with the broader context are important considerations for planning and running general surveillance initiatives. Changes or weaknesses within one component are likely to have implications elsewhere in a general surveillance initiative. These implications could be easily overlooked and they may be sources of significant transaction costs in terms of time, effort and expenditure.

Published: 20 August 2019

Marine pests can cause significant negative social, ecological and economic impacts to infrastructure, marine habitats, water quality, marine industries and coastal amenity values. Maintaining an effective marine pest biosecurity system that minimises the risk of marine pests to Australia is a priority for the Australian Government. The Department of Agriculture commissioned ABARES to investigate the current state of Australia's marine pest biosecurity stakeholder network by means of a social network analysis.

Key Issues

The findings of the study provide a broad understanding of the current marine pest stakeholder network by identifying key players in the network and relationships, and patterns of interaction, between them. The study showed that involvement and interest in marine pest biosecurity is extensive and complex. A wide range of government and non-government organisations and groups participate in the network. The analysis identified opportunities to tap into existing stakeholder networks and build on current structures to further improve network function.

Published: 18 April 2019

The International Convention for the Control and Management of Ships' Ballast Water and Sediments, came into force on 8 September 2017. The Convention regulates the treatment of ballast water to address concerns that ballast water facilitates the spreading of harmful aquatic organisms and pathogens. This report explores the potential impact that the discharge of chemically treated ballast water could have on the water quality of Australian ports.

Key Issues

- It was found that the environmental concentrations of most by-products of chemical treatment were likely to be below levels that would warrant a management response.

- Three compounds were found to potentially exceed environmentally safe levels: dibromoacetonitrile, monochloroacetic acid and dibromoacetic acid.
- The findings are based on modelling, so while they are a guide to possible outcomes the report recommends a physical sampling plan be implemented to determine the actual concentrations of discharge chemicals in Australian ports resulting from the use of ballast water management systems.

Published: 15 November 2017

This report provides details of the method used to develop a database of exotic invasive species that have the potential to have environmental impacts in Australia. The report also provides an overview of species contained in the database.

Three reports published: 28 April 2014; 27 July 2015.

Wild dogs are a significant pest animal in Australia and have a significant detrimental effect on the agricultural sector, but also cause adverse social impacts and are perceived to cause environmental damage.

ABARES examined these impacts in three studies using a mix of economic analysis, interviews with wild dog management groups and a national survey of landholders. The findings from the three studies can inform planning of future investments and development of strategies in wild dog management.

Published: 2 May 2017

The Pest animal and Weed Management Survey: National landholder survey results report presents the key results from a national survey of 6470 agricultural land managers undertaken by ABARES in 2016 about pest and weed management on their property and local area.

This report presents results on a range of topics from the survey including:

- level of awareness of pest animals and Weeds of National Significance (WoNS)
- impacts of pest animals and weeds
- pest animal and weed management activities on the property and in the local area
- and information sources and participation in local support networks.

Published: 18 October 2013.

Prioritising Targets for Biological Control of Weeds - a Decision Support Tool for Policy Makers outlines a framework for prioritising weed targets for biological control and includes a decision support tool that enables policy makers to determine whether biological control is a suitable option for a proposed target species.

Published: 2 March 2012

The proposed biosecurity engagement framework is to provide guidance, insights, tips and tools to conduct effective community engagement for biosecurity purposes. This includes creating an enabling environment for biosecurity engagement (through the proposed national action plan) and providing

principles, tips and tools to engagement practitioners operating in a regional and local context (best recommended practice, tools and mechanisms documents).

This page presents a sample of findings from two ABARES research projects undertaken to examine various social aspects of weed management.

Systematic review of Australian weed-related social surveys

This report, produced by ABARES for the Rural Industries Research and Development Corporation, presents the results of a review of Australian social survey research related to weeds. The social surveys reviewed in the report typically investigate factors such as stakeholders' perceptions of weeds; what they do to address weed issues; what encourages or hinders them in taking action; and where they obtain weed-related information. The report analyses and discusses previously published research of this kind, the questions used in surveys, and the views of relevant experts about this research, and what directions future research should take.

Who's involved with Weeds: Social Network Analysis of funding and information networks for weed management

This report, produced by ABARES for the Rural Industries Research and Development Corporation, examines the funding and information relationships existing between different stakeholders involved in weed management in Australia. This research employed a social network analysis approach to investigate where community groups and institutions (such as local and state government agencies) obtain weed management information and funding.

Source: <https://www.agriculture.gov.au/abares/research-topics/biosecurity/biosecurity-sciences>